

Math 242, S16
Exam 1

Name: *Answer*
SSN: xxx-xx-

Instructions:

- There are totally 5 problems.
- You must show all of your work to receive a credit for a correct answer.

Problem	Points	Score
1	24	
2	24	
3	16	
4	16	
5	20	
Total	100	

Good Luck !

Problem 1. (24 points) Solve the following initial value problems:

(a) $\frac{dy}{dx} = 2xy^4, \quad y(1) = 1.$

$$\begin{aligned} \frac{1}{y^4} dy &= 2x dx \\ \int \frac{1}{y^4} dy &= \int 2x dx & \Rightarrow \frac{1}{\sqrt[3]{-3x^2 + C}} = 1 \\ -\frac{1}{3} y^{-3} &= x^2 + C, & -3 + C = 1 \\ y^{-3} &= -3x^2 + C & C = 4 \\ y &= \frac{1}{\sqrt[3]{-3x^2 + C}} & \text{Thus } y(x) = \frac{1}{\sqrt[3]{-3x^2 + 4}} \quad (\text{or } (-3x^2 + 4)^{-\frac{1}{3}}) \\ & \quad (\text{or } (-3x^2 + C)^{-\frac{1}{3}}) \end{aligned}$$

(b) $(1+x^2)\frac{dy}{dx} - 2x(1+x^2)y = 2xe^{x^2}, \quad y(0) = 2.$

Standard form: $\frac{dy}{dx} - 2xy = \frac{2xe^{x^2}}{1+x^2}$

$(P(x) = -2x, Q(x) = \frac{2xe^{x^2}}{1+x^2})$

Integrating factor: $\int P(x) dx = \int -2x dx = -x^2$

Thus $\frac{d}{dx} [y e^{-x^2}] = \frac{2xe^{x^2}}{1+x^2} (e^{-x^2})$

$y e^{-x^2} = \int \frac{2x}{1+x^2} dx$

$y e^{-x^2} = \ln(1+x^2) + C$

$y = e^{x^2} (\ln(1+x^2) + C)$

$y(0) = 2 \Rightarrow e^0 (\ln 1 + C) = 2 \Rightarrow C = 2$

Thus $y(x) = e^{x^2} (\ln(1+x^2) + 2)$

Problem 2. (24 points) Find a general solution for the following differential equations:

(a) $\frac{dy}{dx} = (x + 4y)^2$ (Hint: use an appropriate substitution)

Let $V = x + 4y$, then $y = \frac{1}{4}V - \frac{1}{4}x$ and $\frac{dy}{dx} = \frac{1}{4}\frac{dV}{dx} - \frac{1}{4}$

$$\Rightarrow \frac{1}{4}\frac{dV}{dx} - \frac{1}{4} = V^2$$

$$\arctan(2V) = 2X + C$$

$$\frac{dV}{dx} = 1 + 4V^2$$

$$2V = \tan(2X + C)$$

$$V = \frac{1}{2}\tan(2X + C)$$

$$\frac{1}{1+4V^2}dV = dx$$

$$\int \frac{1}{1+4V^2}dV = \int dx$$

$$\Rightarrow y = \frac{1}{4}V - \frac{1}{4}x = \frac{1}{8}\tan(2X + C) - \frac{1}{4}x$$

$$\frac{1}{2}\arctan(2V) = X + C_1$$

(b) $(3x^2 + 2y^2)dx + (4xy + 6y^2)dy = 0$ (Hint: this is an exact equation)

$$\begin{aligned} M(x, y) &= 3x^2 + 2y^2 \Rightarrow \frac{\partial M}{\partial y} = 4y \\ N(x, y) &= 4xy + 6y^2 \Rightarrow \frac{\partial N}{\partial x} = 4y \end{aligned} \rightarrow \text{exact!}$$

$$\text{Thus } \frac{\partial F}{\partial x} = M, \frac{\partial F}{\partial y} = N$$

$$\Rightarrow F(x, y) = \int M(x, y)dx = \int (3x^2 + 2y^2)dx = x^3 + 2xy^2 + g(y)$$

$$\frac{\partial F}{\partial y} = N \Rightarrow \frac{\partial}{\partial y}(x^3 + 2xy^2 + g(y)) = 4xy + 6y^2$$

$$4xy + g'(y) = 4xy + 6y^2$$

$$g'(y) = 6y^2$$

$$\Rightarrow g(y) = \int 6y^2 dy = 2y^3 + C$$

$$\text{Thus } F(x, y) = x^3 + 2xy^2 + 2y^3$$

and the general solution is

$$x^3 + 2xy^2 + 2y^3 = C$$

Problem 3. (16 points) A certain city has a population of 20000 initially and a population of 30000 after 20 years. Assume that its population grows at a constant rate k per year. Find the population of the city $P(t)$ and how long it takes for the population to reach 40000?

$$\begin{aligned}\frac{dP}{dt} &= kP, & \text{We solve} \\ \Rightarrow P(t) &= A e^{kt} & P(t) &= 40000 \\ P(0) &= 20000 \Rightarrow A = 20000 & \Rightarrow 20000 e^{\frac{t}{20} \ln(\frac{3}{2})} &= 40000 \\ \Rightarrow P(t) &= 20000 e^{kt} & e^{\frac{t}{20} \ln(\frac{3}{2})} &= 2 \\ P(20) &= 30000 & t \left(\frac{1}{20} \ln(\frac{3}{2}) \right) &= \ln 2 \\ & & t &= \frac{20 \ln 2}{\ln(\frac{3}{2})} \approx \underline{34.19 \text{ years}} \\ & \Rightarrow 20000 e^{20k} = 30000 & & \\ & e^{20k} = \frac{3}{2} & & \\ & 20k = \ln(\frac{3}{2}) & & \\ & k = \frac{1}{20} \ln(\frac{3}{2}) \approx 0.02027 & \Rightarrow P(t) &= 20000 e^{\frac{t}{20} \ln(\frac{3}{2})}\end{aligned}$$

Problem 4. (16 points) Given an initial value problem

$$\frac{dy}{dx} = y - 2x + 1, \quad y(0) = -1.$$

Apply **Euler's method** to approximate the solution of the initial value problem on the interval $[0, 0.3]$ with step size $h = 0.1$.

$$\begin{aligned}x_0 &= 0, \quad y_0 = -1, \quad h = 0.1, \quad f(x, y) = y - 2x + 1 \\ x_1 &= x_0 + h = 0 + 0.1 = 0.1 \\ y_1 &= y_0 + h f(x_0, y_0) = y_0 + h [y_0 - 2x_0 + 1] \\ &= -1 + 0.1 \times [-1 - 2 \cdot 0 + 1] = -1 + 0 = -1 \\ x_2 &= x_1 + h = 0.1 + 0.1 = 0.2 \\ y_2 &= y_1 + h f(x_1, y_1) = y_1 + h [y_1 - 2x_1 + 1] \\ &= -1 + 0.1 \times [-1 - 2 \times 0.1 + 1] = -1 + (-0.02) = -1.02 \\ x_3 &= x_2 + h = 0.2 + 0.1 = 0.3 \\ y_3 &= y_2 + h f(x_2, y_2) = y_2 + h [y_2 - 2x_2 + 1] \\ &= -1.02 + 0.1 \times [-1.02 - 2 \times 0.2 + 1] = -1.02 + 0.1 \times (-0.42) \\ &= -1.02 - 0.042 = -1.062\end{aligned}$$

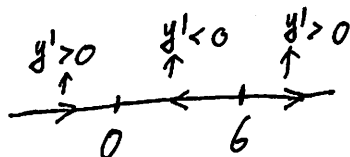
Problem 5. (20 points) Given the following initial value problem:

$$\frac{dy}{dt} = 3y^2 - 18y, \quad y(0) = 5.$$

(a) Find all critical points of the given autonomous differential equation and then determine whether it is stable or unstable for each of them.

$$3y^2 - 18y = 3y(y-6) = 0$$

$$\Rightarrow y = 0 \text{ or } y = 6 \quad \text{two critical points}$$



$$y = 0 \quad \text{stable}$$

$$y = 6 \quad \text{unstable}$$

(b) Solve the initial value problem for $y(t)$.

$$\frac{dy}{dt} = 3y(y-6)$$

$$\int \frac{1}{y(y-6)} dy = \int 3 dt$$

$$\int \left(\frac{1}{y} - \frac{1}{y-6} \right) dy = \int 3 dt$$

$$\int \left(\frac{1}{y} - \frac{1}{y-6} \right) dy = \int -18 dt$$

$$\ln|y| - \ln|y-6| = -18t + C$$

$$\ln \left| \frac{y}{y-6} \right| = -18t + C$$

$$\left| \frac{y}{y-6} \right| = e^{-18t+C}$$

$$\frac{y}{y-6} = Be^{-18t}$$

$$y(0) = 5 \Rightarrow \frac{5}{5-6} = Be^0$$

$$\Rightarrow B = -5$$

$$\frac{y}{y-6} = -5e^{-18t}$$

$$(-5e^{-18t})y + 30e^{-18t} = y$$

$$(1 + 5e^{-18t})y = 30e^{-18t}$$

$$y = \frac{30e^{-18t}}{1 + 5e^{-18t}}$$

$$= \frac{30}{5 + e^{18t}}$$